

O10e “Michelson Interferometer”

Tasks

1. Adjust a Michelson interferometer and determine the wavelength of a He-Ne laser.
2. Measure the change in the length of a piezoelectric actor when a voltage is applied. Plot the length change as a function of voltage and determine the sensitivity of the sensor.
3. Measure the dependence of the refractive index of air as a function of the air pressure p . Plot $\Delta n(p)$ and calculate the index of refraction n_0 at standard conditions.
4. Measure the length of a ferromagnetic rod as a function of an applied magnetic field. Plot the relative length change versus the applied field.
5. Determine the relative change in the length of a metal rod as a function of temperature and calculate the linear expansion coefficient.

Literature

Physics, P.A. Tipler 3. Ed., Vol. 2, Chap. 33-3

University Physics, H. Benson, Chap. 37.6

Physikalisches Praktikum, 13. Auflage, Hrsg. W. Schenk, F. Kremer, Optik, 2.0.1, 2.0.2, 2.4

Accessories

He-Ne laser, various optical components for the setup of a Michelson interferometer, piezoelectric actor with mirror, laboratory power supply, electromagnet, ferromagnetic rod with mirror, metal rod with heating filament and mirror, vacuum chamber with hand pump.

Keywords for preparation

- Interference, coherence
- Basic principle of the Michelson interferometer
- Generation and properties of laser light
- Index of refraction, standard conditions
- Piezoelectricity, magnetostriction, thermal expansion

In this experiment you will work with high quality optical components. Work with great care! While operating the LASER do not look directly into the laser beam or its reflections!

Basics

The time and position dependence of a plane wave travelling in the positive (negative) z-direction is given by

$$\psi = \psi_0 \exp[i(\omega t \mp kz + \varphi)], \quad (1)$$

where ψ might denote e.g. a component of the electric field. The phase of the wave is given by the expression $(\omega t \mp kz + \varphi)$, the wave vector by $k = 2\pi/\lambda$, the angular frequency by $\omega = 2\pi/T$. φ denotes the phase angle at zero. The phase velocity of the wave, v_{ph} , which in vacuum is equal to the velocity of light c_0 , follows from the condition of constant phase to $dz/dt = v_{ph} = \omega/k$.

Interference occurs, when two (or more) wave fronts superimpose at a given position. During superposition the resulting wave is either amplified or weakened, depending on the relative phases. The most important precondition for the observability of interference is that the superimposing waves are coherent, i.e. have a fixed phase relation.

The phase difference of two waves is – taking the presence of two media 1 and 2 into account – given by $\Delta\Phi = (k_1\Delta z_1 + \varphi_1) - (k_2\Delta z_2 + \varphi_2)$, where Δz_1 and Δz_2 are the geometric path differences along which the two waves have travelled. If the pairwise occurring phase jumps due to reflection at a fixed end compensate, as is the case in the present experiment, the phase difference is caused by a difference in kz or nz , respectively. The latter quantity is called the optical path length (n : refractive index). The optical path difference is then given by $\Delta = n_1\Delta z_1 - n_2\Delta z_2 = (\Delta\Phi/2\pi)\lambda_0$, where λ_0 denotes the vacuum wave length.

When two waves $\psi_1(z, t)$ and $\psi_2(z, t)$ with phase difference $\Delta\Phi$ superimpose, the resulting intensity is given by $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(\Delta\Phi)$. One obtains constructive interference for $\Delta\Phi = 2n\pi$ and destructive interference for $\Delta\Phi = (2n+1)\pi$.

For the considerations so far, ideal, i.e. monochromatic as well as infinite waves in both time and space, were assumed. However, every wave, as well as every spectral line, has due to its finite bandwidth Δf (resp. spectral width $\Delta\lambda$) a finite length of its wave train, namely the coherence length L_k , or equivalently, the coherence time $\Delta t = L_k/c_0$. The uncertainty principle for waves leads to $\Delta k \Delta z \approx 1/2$ and $\Delta f \Delta t \approx 1/(4\pi)$. Therefore one has

$$L_k \approx \frac{1}{2\Delta k} \approx \frac{\lambda_0^2}{4\pi\Delta\lambda}. \quad (2)$$

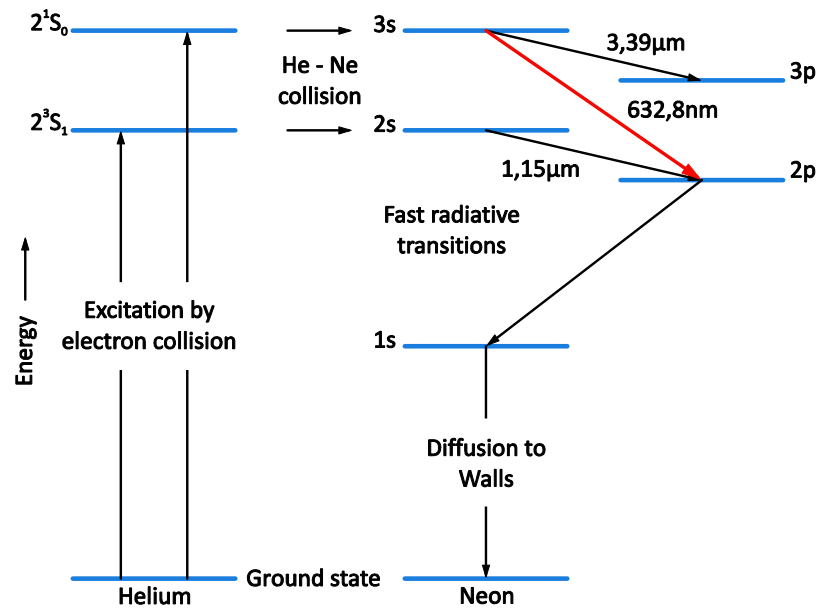
The finite coherence length determines the maximally possible optical path difference in interference instruments.

In many setups the coherence of the superimposing waves is generated by beam splitting using half-transparent mirrors or other devices. This does only work in case of spatially extended light sources (with a typical spatial extension of a), if the coherence condition is fulfilled (θ denotes the angle of the beam spread):

$$2a \sin(\theta) \ll \lambda/2. \quad (3)$$

Laser light is especially coherent with coherence lengths of the order of 1 m. It is generated by stimulated emission. The excited atoms emit the light frequency – that is determined by the mirror distance – coherently with equal phases. In the He/Ne gas mixture an electric discharge is sustained between two electrodes and the He atoms are excited by electron collisions. The excitation of the Ne atoms into the relevant 3s state occurs by He-Ne collisions with simultaneous energy transfer (collisions of the second kind). At standard operation the He-Ne-LASER emits light with a wavelength $\lambda = 632.8 \text{ nm}$.

Fig. 1 He-Ne energy diagram



Adjustment

The He-Ne laser beam is split by a half-transparent mirror (Fig. 2 **b**, in our setup a beam-splitter cube is used) into two partial beams of approximately the same intensity. One resultant beam is guided towards the planar mirror **d** and will be reflected from there through the beam divider directly onto the screen **h**. The second beam is reflected by the adjustable mirror **c** (coupled with the fine-adjustment mechanism) and arrives at the screen **h** after being reflected again by the beam divider **b**. The diffraction pattern is clearly visible, if the two beams superimpose, if the surfaces of the optical components are rather flat and if the optical path lengths of the two beams do not differ more than half the coherence length from each other. A spherical lens **e** is used for a better observation of the diffraction pattern. If the setup is well adjusted, a diffraction pattern as shown in Fig. 3 should be observable on the screen.

The width and quality of the line fringes are dependent on the setup. The better the adjustment, the wider are the fringes. This is due to the fact that a sloppy adjustment leads to a larger optical path difference between the two interfering rays and therefore to a bigger number of interference fringes in the interference pattern. A very dense diffraction pattern is not of advantage, because counting the number of light-dark fringes is more difficult. When the optical path length is continuously changed, the fringes should move slowly across the screen.

Fig. 2 Michelson interferometer with fine adjustment drive

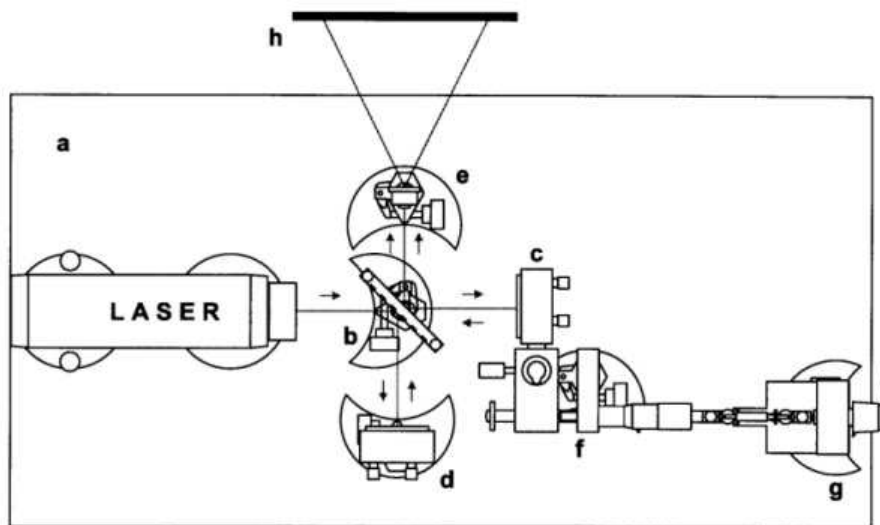
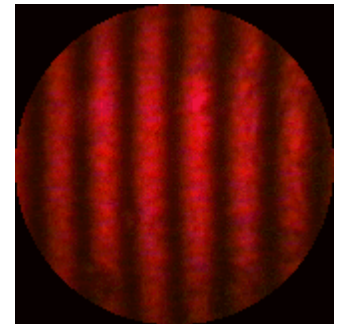


Fig. 3 Interference pattern with straight line fringes



The experimental setup is quite sensitive against vibrations. Therefore the optical components are put up on a massive socket - base plate (**a**) in Fig. 2 – which is seated on a pneumatic damping-system in order to damp the vibrations of the table. The base plate is relatively small, so the setup has to be well planned. In a slightly darkened room assemble and adjust the optical components as follows:

- Attach the aperture to the opening of the laser beam, switch on laser.
- Adjust the laser mount, so that the beam path is parallel to the surface of the base plate.
- Adjust the beam divider and the mirrors using the fine adjustment screws, such that the laser beam is reflected back near the laser aperture. The quality of the laser beam is affected, if the reflected split beams fall back right into the aperture of the laser.
- Setup beam divider (**b**) according to Fig. 2, such that one partial beam is reflected at an angle of approximately 90 degrees with respect to the original beam direction.
- Setup the fixed mirror (**d**) in the beam path, such that the reflected beam hits the beam divider in the same spot as the original laser beam.
- Setup the adjustable mirror (**c**), such that the reflected beam hits the beam divider (**b**) in the same spot as the primary laser beam leaves the half-mirror.
- Position screen (**h**) as shown in Fig. 2, but without the spherical lens with antireflective coating (**e**, focal length $f=2.7$ mm). Turn the adjustable mirror (**c**), such that the split beams superimpose on the screen and that also the points appearing on the half mirror will remain on top of each other. Further partial beams with low intensity will appear beside the major rays; these are due to multiple reflections and will be removed later by the lens holder.
- Place the spherical lens (**e**), such that the overlaid beams hit the lens opening. At the same time check the diffraction pattern. The fringes should not be too wide or too narrow for a sensitive measurement. If necessary, readjust the beams.

The micrometer screw which is connected to the fine adjustment mechanism is turned by an electronic motor for easier measurement. The motor voltage supply (1V to 6V) for the engine is supplied by a laboratory voltage supply. The micrometer screw has a gear reduction ratio 100:1 and moves the mirror a well-defined length. The direction of rotation of the motor is equivalent to the

direction of rotation of the micrometer screw. During the measurement the micrometer screw should act against the pressure of the spring and the sense of rotation of the motor has to be chosen accordingly by the polarity of the voltage. The measurement is started, when the movement of the fringes is continuous. Count the passing fringes using a mark on the screen for at least 10 rotations of the motor.

For the determination of the wavelength use the following equation:

$$\lambda = 2 \frac{0.5 \cdot 10^{-3}}{100} \frac{k}{z} \text{ (m)} \quad (4)$$

k : number of turns of the micrometer screw, z : number of fringes, $0.5 \cdot 10^{-3}$: scaling factor relating one turn of the micrometer screw to the displacement of the mirror.

Measurements with a piezoelectric sensor

The adjustable mirror connected to the micrometer screw and motor is exchanged with the piezoelectric sensor. Readjust the interferometer. Pay attention to the correct polarity of the voltage supply. *In principle, a maximum positive voltage of 150 DCV might be applied to the piezoelectric actor, but small negative voltages of -15 DCV might already destroy it!*

The voltage for the piezoelectric sensor is supplied by a laboratory power supply in the range between 0 and 40 DCV. Measure the hysteresis of the piezoactor for increasing and decreasing voltages. Plot the length change Δl versus the voltage and determine the average slope of the graph (in units of $\mu\text{m/V}$).

Measurement of the index of refraction of air

Place the test chamber in one of the beam paths such that the partial beam passes through it axially. Use the hand pump with pressure display to evacuate the chamber. Carry out about eight measurements between atmospheric air pressure and a chamber pressure of approximately 100 hPa. Determine the number z of fringes passing a mark on the screen in dependence on the pressure inside the chamber. Use the following equations for the determination of the index of refraction of air at the atmospheric pressure $p = p(\text{air})$.

$$s_{\text{opt}} = \int n \cdot ds_{\text{geom}} \quad \begin{array}{l} n : \text{ index of refraction} \\ s_{\text{opt}} : \text{ optical path length} \\ s_{\text{geom}} : \text{ geometric path length} \\ d = 50 \text{ mm (length of the test chamber)} \\ s_{\text{geom}} = 2d \text{ (Michelson interferometer)} \end{array}$$

$$n(p) - n(p=0) = \frac{\Delta n}{\Delta p} p, \quad n(p=0) = 1 \text{ (vacuum refractive index)}, \quad \Delta n = \frac{\Delta s_{\text{opt}}}{s_{\text{geom}}} = \frac{\Delta z \lambda}{s_{\text{geom}}}$$

$$n(p) = 1 + \frac{\Delta z}{\Delta p} \cdot \frac{\lambda}{s_{\text{geom}}} \cdot p \quad (5)$$

with $p = p(\text{air})$. Determine the slope $\Delta z / \Delta p$ graphically. Calculate the index of refraction of air at standard conditions. Using the relationship $n-1 \propto N$ (N : number of particles per volume,

$N = N_A p / RT$ assuming an ideal gas, $\gamma = \frac{1}{273.15} \text{K}^{-1}$ one obtains $(n-1) = (n_0-1) \frac{p}{p_0(1+\gamma\vartheta)}$. (Derive this equation!)

Measurement of the magnetostriction of a ferromagnetic rod

For the study of magnetostriction a ferromagnetic rod with given length l is used, with a plane mirror attached to one end. The rod is installed instead of the mirror (c). Place a coil around the rod which creates a magnetic field that in turn leads to a length change Δl of the rod. Coil current should be in the range $I = 0 \dots 2.5 \text{ A}$. In order to avoid strong heating of the coil, the measurements at high currents should be speedily made. To determine the sign of the length change (extension or contraction) the rod is subsequently warmed with a blow-dryer. If the interference fringes move on warming in the same direction as on increase of the coil current, the magnetostriction is positive, i.e. an extension; otherwise it is negative (contraction). Plot the relative length change $\varepsilon = \Delta l / l = f(B)$, where B is the magnetic induction measured with a Teslameter. In order to obtain reproducible values, it would be favourable to demagnetize the rod in a magnetic ac field of decreasing amplitude, before the start of the experiment.

Measurement of the linear thermal expansion

In this case a metal rod with integrated heater foil is used; again a plane mirror is attached to one end of the rod. The heater current is supplied by a laboratory power supply and the temperature of the rod is measured with an integrated thermal sensor. The thermal expansion coefficient α of the rod (of known length l) can be determined under the assumption of a linear thermal expansion from

$$\alpha = \frac{\lambda}{2l} \frac{\Delta z}{\Delta T}. \quad (6)$$

The rod should be warmed up to a maximum temperature of 60°C with a maximum heater power of 15 W . Subsequently, the length change is measured on cooling from about 45°C down to room temperature. In the experiment the interference fringes moving across a mark are counted and the dependence $z(T)$ is plotted. $\Delta z / \Delta T$ is determined from the slope of the graph. The length of the rod l is measured with a ruler.